

Types of Operating Systems

1 Single-User Operating System

A **single-user operating system** is designed to allow **only one user to use the computer at a time**. It manages the computer's hardware and software resources for that user.

Example: MS-DOS.

Scenario in daily life: A personal computer where one person logs in and uses the system.

2 Multiprogramming Operating System

A **multiprogramming operating system** allows **several programs to be kept in the main memory at the same time**. When one program is waiting for an input or output operation, the CPU can work on another program. This helps to keep the CPU busy and improves system efficiency.

Example: UNIX.

Scenario in daily life: If one program is waiting for a file to load, the CPU can process another program instead of remaining idle.

3 Multitasking Operating System

A **multitasking operating system** allows a user to **run and work with several applications at the same time**. The operating system quickly switches the CPU between different tasks, making them appear to run simultaneously.

Example: Windows 10/11, macOS.

Scenario in daily life: A user can listen to music, browse the internet, and type a document at the same time.

4 Multiprocessing Operating System

A **multiprocessing operating system** supports the use of **two or more processors or CPU cores** to perform tasks. Different processors or cores can execute different tasks simultaneously, which increases the speed and performance of the computer.

Example: Linux, Windows, and macOS running on multi-core computers.

Scenario in daily life: One CPU core can process a web browser while another core handles a video player at the same time.

5 Real-Time Operating System

A **Real-Time Operating System (RTOS)** is an operating system designed to respond to events or requests within a **specific time limit**. It is mainly used in systems where timely response is important.

There are two main types of real-time operating systems:

5.1 Hard Real-Time Operating System

A **Hard Real-Time Operating System** must complete a task within a strict deadline. Missing the deadline can cause serious problems or failure of the system.

Example: Airbag control systems, aircraft control systems, and medical equipment.

Scenario in daily life: An airbag system must respond immediately when a crash is detected. A delayed response could be dangerous.

5.2 Soft Real-Time Operating System

A **Soft Real-Time Operating System** also tries to complete tasks within a given time limit, but occasional delays are acceptable. Missing a deadline may reduce performance but usually does not cause complete system failure.

Example: Video streaming, online gaming, and multimedia systems.

Scenario in daily life: If a video frame is slightly delayed while watching a video, the video may become less smooth, but the system can continue working.

6 Distributed Operating System

A **distributed operating system** manages a group of independent computers and makes them work together as if they were a **single system**. The computers communicate with each other through a network and share resources and tasks.

Example: Distributed systems used in large computing networks and research environments.

Scenario in daily life: A large task can be divided among several connected computers. Each computer performs part of the task, and the results are combined to complete the overall task.

7 Difference Between Single-User, Multiprogramming, Multitasking, and Multiprocessing OS

Single-User OS	Multiprogramming OS	Multitasking OS	Multiprocessing OS
Allows one user to use the system at a time.	Keeps multiple programs in memory at the same time.	Allows a user to run multiple tasks at the same time.	Uses two or more processors or CPU cores.
Focuses mainly on supporting a single user.	Focuses on keeping the CPU busy by switching between programs.	Focuses on allowing multiple tasks to run at the same time.	Focuses on executing multiple tasks simultaneously.
Usually handles one user's activities.	The CPU switches to another program when one program is waiting.	The CPU rapidly switches between different tasks.	Different processors or cores can execute different tasks.
It may have limited resource-sharing capabilities.	Improves CPU utilization and reduces CPU idle time.	Improves user convenience and system responsiveness.	Improves performance and processing speed.
Example: MS-DOS	Example: UNIX	Example: Windows 11	Example: Linux on a multi-core computer

Chapter 3.3: Windows Operating System

1 User Interface

A **User Interface (UI)** is the way through which a user communicates with and controls a computer. The two common types of user interfaces are **CUI/CLI** and **GUI**.

1.1 CUI / CLI (Character/Command Line User Interface)

CUI (Character User Interface) or **CLI (Command Line Interface)** allows the user to interact with the computer by **typing commands using the keyboard**.

For example, in Command Prompt, a user can type:

```
dir
```

to view files and folders.

Features of CUI/CLI

- Uses text-based commands.
- Mainly uses the keyboard.
- Requires knowledge of commands.
- Uses very few graphical elements.
- Can be fast for experienced users.
- Requires less memory and processing power.

Examples: MS-DOS and Windows Command Prompt.

1.2 GUI (Graphical User Interface)

GUI (Graphical User Interface) allows users to interact with the computer using **graphical elements** such as icons, menus, buttons, and windows.

For example, in Windows, a user can double-click an icon to open a program.

Features of GUI

- Uses icons, windows, menus, and buttons.
- Mainly uses the mouse and keyboard.
- Easy to learn and use.
- Does not require memorizing many commands.
- Provides a visual and user-friendly environment.
- Supports point-and-click operations.

Examples: Microsoft Windows and macOS.

1.3 Difference Between CUI/CLI and GUI

Basis	CUI/CLI	GUI
Interaction	Uses typed commands	Uses graphical elements
Input	Mainly keyboard	Mouse and keyboard
Ease of use	Difficult for beginners	Easy for beginners
Commands	Commands must be remembered	Commands are selected using menus and icons
Appearance	Text-based	Graphical and visual

2 Introduction to Windows OS

Windows is a popular **graphical operating system** developed by **Microsoft** that provides an easy-to-use environment for running programs, managing files and folders, and controlling computer hardware.

Some common versions of Windows include **Windows 7, Windows 10, and Windows 11**.

2.1 Features of Windows

Some important features of Windows are:

1. Provides a graphical and user-friendly interface.
2. Allows multiple applications to run at the same time.
3. Helps users create, organize, copy, move, and delete files.
4. Supports devices such as printers, keyboards, and USB drives.
5. Provides features such as passwords, user accounts, and security tools.

2.2 Main Elements of Windows

- **Desktop:** The main screen that appears after Windows starts.
- **Icons:** Small graphical symbols representing programs, files, folders, or other items.
- **Pointer:** The arrow or other symbol controlled by the mouse.
- **Taskbar:** The bar that shows open applications and provides quick access to programs.
- **Windows:** Rectangular areas where applications, files, or folders are displayed.
- **Start Button:** Provides access to applications, settings, files, and other Windows features.

3 Files and Folders

3.1 File

A **file** is a collection of related data stored on a computer with a specific name.

Examples:

- `notes.docx`
- `photo.jpg`
- `song.mp3`

Files are used to store different types of information such as text, images, videos, audio, and programs.

3.2 Folder

A **folder** is a container used to **store and organize files and other folders**.

For example, a folder named **College Notes** may contain different subject files:

```
College Notes
|
+--- Computer.docx
+--- Mathematics.pdf
+--- OperatingSystem.docx
```

Folders make it easier to organize, manage, and find files.

4 File Extensions

A **file extension** is the group of letters after the **dot (.)** in a filename. It tells the operating system what type of file it is and which program can usually open it.

For example:

```
assignment.docx
```

Here, `.docx` is the file extension.

4.1 Common File Extensions

Extension	File Type	Example
.txt	Text file	notes.txt
.docx	Microsoft Word document	report.docx
.pdf	PDF document	book.pdf
.xlsx	Excel spreadsheet	marks.xlsx
.pptx	PowerPoint presentation	presentation.pptx
.jpg / .jpeg	Image	photo.jpg
.png	Image	logo.png
.gif	Image/Animated file	animation.gif
.mp3	Audio file	song.mp3
.mp4	Video file	movie.mp4
.avi	Video file	video.avi
.zip	Compressed file	files.zip
.exe	Executable program	setup.exe
.html	Web page	index.html
.csv	Comma-separated data file	data.csv

1 Open Source Operating System

An **Open Source Operating System** is an operating system whose **source code is freely available** for users and developers to view, modify, and distribute according to its license.

Open source operating systems are usually developed by a community of developers rather than by a single company.

Features of Open Source OS

- Source code is available to users and developers.
- Can be modified and customized.
- Usually has strong community support.
- Developers can find and fix problems.
- Often available free of cost.

Examples: Linux, Ubuntu, and FreeBSD.

1.1 Linux Operating System

Linux is an open-source operating system based on the Linux kernel. It is widely used on computers, servers, embedded systems, and many other devices.

Linux is known for its **security, stability, flexibility, and customization**. Users can modify the system according to their requirements.

Examples of use: Web servers, personal computers, supercomputers, and embedded devices.

Features of Linux

- Open source and highly customizable.
- Secure and stable.
- Supports multitasking and multi-user operation.
- Available in many distributions.
- Widely used for servers and development.

1.2 UNIX Operating System

UNIX is a powerful operating system originally developed at Bell Labs. It was designed to support multiple users and multiple tasks efficiently.

UNIX is widely known for its **stability, security, and powerful command-line environment**. Many modern operating systems and operating-system concepts have been influenced by UNIX.

Features of UNIX

- Supports multiple users.
- Supports multitasking.
- Provides strong security.
- Stable and reliable.
- Provides a powerful command-line interface.

Examples: UNIX systems are used in servers, workstations, and other large computing environments.

1.3 Linux Distribution: Ubuntu

Ubuntu is a popular Linux distribution based on the Linux operating system. It is developed and maintained by Canonical and a large open-source community.

Ubuntu provides a **user-friendly graphical interface** and is suitable for beginners as well as experienced users.

Features of Ubuntu

- Free and open source.
- Easy-to-use graphical interface.
- Secure and stable.
- Supports many software applications.
- Available for desktops and servers.
- Receives regular updates and security fixes.

Example: Ubuntu Desktop can be used as an alternative to Windows for everyday computer activities.

2 Mobile Operating System

A **Mobile Operating System (Mobile OS)** is an operating system designed specifically for **mobile devices** such as smartphones, tablets, and other portable devices.

It manages the device's hardware and provides an environment for running mobile applications.

Features of Mobile OS

- Provides a touch-based user interface.
- Manages mobile hardware and resources.
- Supports mobile applications.
- Provides wireless communication and networking.
- Manages features such as cameras, GPS, and sensors.
- Provides security and user privacy features.

3 Types of Mobile Operating Systems

3.1 Android OS

Android is a mobile operating system developed primarily by Google and based on the Linux kernel. It is widely used in smartphones, tablets, TVs, and other smart devices.

Example: Samsung, Google Pixel, and Xiaomi smartphones.

3.2 iOS

iOS is a mobile operating system developed by Apple for its mobile devices. It is known for its simple interface, security, and integration with Apple's hardware and services.

Example: iPhone.

3.3 Palm OS

Palm OS was a mobile operating system originally developed for personal digital assistants (PDAs). It was designed to provide a simple and easy-to-use interface for portable devices.

Example: Palm handheld devices.

3.4 BlackBerry OS

BlackBerry OS was a mobile operating system developed by BlackBerry for its smartphones. It was particularly known for its **secure communication and physical QW-ERTY keyboards**.

Example: BlackBerry smartphones.

3.5 Symbian OS

Symbian OS was a mobile operating system widely used in early smartphones. It was especially popular in Nokia mobile phones before the widespread adoption of Android and iOS.

Example: Nokia smartphones.

3.6 HarmonyOS

HarmonyOS is an operating system developed by Huawei. It is designed for smartphones and a wider range of connected and smart devices.

HarmonyOS focuses on providing a connected experience across different types of devices.

Example: Huawei smartphones and other Huawei smart devices.